26900424 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECPHYA00 - RESEARCH METHODOLOGY, SCIENTIFIC WRITING AND LATEX FOR PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ with options

Answer all questions

Each question carries **1** marks

1. Quantitative analysis commonly uses [A] / [CO2]

   a). Personal opinions | b). Statistical techniques and numerical methods | c). Descriptive narratives only | d). Unstructured observations

2. An unanswered need can serve as [U] / [CO1]

   a). A final research result | b). A basis for a research problem | c). A research conclusion | d). A literature review

3. Replicating previous research with modifications can [An] / [CO1]

   a). Eliminate the need for further research | b). Generate new or unexpected findings | c). Prove that the original research was incorrect | d). Avoid reviewing previous literature

4. A numerical simulation of particle motion is an example of [An] / [CO1]

a). Computational research b). Experimental Research c). Qualitative Research d). Theoretical Research

5. Both basic and applied research can contribute to [A] / [CO1]

a). Only theoretical knowledge b). Advancement of knowledge c). Only commercial development d). Only laboratory techniques

6. Which approach can complement both experimental and theoretical work? [U] / [CO1]

a). Experimental approach b). Theoretical research c). Computational research d). Qualitative Research

7. A good research question should be [A] / [CO1]

a). Vague and broad b). Clear, focused and answerable c). Based only on personal opinion d). Impossible to investigate

8. Which statement is most precise about delimiting a research problem? [U] / [CO1]

a). It eliminates the need for research objectives.

b). It allows the researcher to study every aspect of the topic.

c). It makes the research topic broader.

d). Defines specific boundaries, scope, population and variables of the study.

9. Precision refers to [A] / [CO2]

a). Closeness of repeated measurements to one another

b). The closeness of a measurement to the true value

c). The size of the measuring instrument

d). The number of variables in an experiment

10. Which is NOT a characteristic of a good hypothesis? [U] / [CO1]

a). Clear and specific b). Testable and measurable c). Based on existing knowledge or theory d). It cannot be tested

11. The literature review is important because it [A] / [CO3]

a). Replaces the need for data collection b). Places the study in the context of existing knowledge c). Guarantees the research results d). Eliminates the need for analysis

12. A direct quotation in a research report should be: [U] / [CO3]

a). Used without mentioning the source

b). Rewritten without acknowledging the original author

c). Presented without quotation marks or citation

d). Clearly indicated as quoted text and properly cited with the original source

13. A good interpretation of research results considers: [A] / [CO3]

a). Only the numerical values obtained

b). Only the researcher's personal opinion

c). The findings, evidence, research objectives, and relevant previous studies

d). Information unrelated to the research

14. Research ethics helps protect: [K] / [CO3]

a). Only the reputation of research institutions

b). Only the financial interests of researchers

c). The integrity and credibility of research

d). Only the published research papers

15. Good presentation of findings may use [An] / [CO3]

a). Only long paragraphs of text

b). Only personal opinions

c). Unorganized raw data

d). Words, tables and graphics

16. The introduction of a final report should establish [A] / [CO3]

a). The background and research problem

b). Only the final results

c). The reference list

d). The appendix

17. A long quotation in academic writing should generally be: [A] / [CO3]

a). Placed in quotation marks within every sentence

b). Indented and presented separately

c). Included without quotation marks or citation

d). Paraphrased without acknowledging the original source

18. A researcher should identify sub-areas before [U] / [CO3]

a). Writing the conclusion

b). Formatting the references

c). Compiling the report

d). Conducting an extensive literature review

19. A proposal contains a literature review to [U] / [CO3]

a). Provide only a list of references related to the topic

b). Describe the detailed findings of the completed experiment

c). Present the final results obtained from the research

d). Establish the existing knowledge, identify research gaps, and justify the proposed study

20. The introduction of a literature review should describe [A] / [CO3]

a). The reference list

b). The final results of the study

c). The background and context of the research topic

d). Only the researcher's personal opinions

21. The purpose of compiling is to [K] / [CO4]

a). Convert source code into formatted output

b). Collect research data

c). Insert references manually

d). Design experimental equipment

22. Automatic numbering is an advantage because it [A] / [CO4]

a). Makes manual editing necessary

b). Removes the need for headings

c). Prevents the use of references

d). Maintains consistency when content changes

23. Scientific illustrations can be included using [A] / [CO4]

a). `\caption{}`

b). `\maketitle{}`

c). `\label{}`

d). Figure environment with `\includegraphics{}`

24. The LaTeX command for italic text is: [A] / [CO4]

a). `\textbf{}`

b). `\textit{}`

c). `\texttt{}`

d). `\underline{}`

25. A citation should allow readers to [U] / [CO4]

a). Avoid reading the research paper

b). Change the research methodology

c). Verify the researcher’s calculations

d). Identify the original source

26. A table is appropriate for [A] / [CO4]

a). Showing trends over time only | b). Writing lengthy explanations | c). Describing personal opinions | d). Organized experimental observations

27. The tabular environment specifies [K] / [CO4]

a). The title of a document | b). The font size of a paragraph | c). The page margins | d). The structure and allignment of table

28. The letter class in LaTeX is used for: [A] / [CO4]

a). Creating presentation slides | b). Preparing formal letters and correspondence

c). Writing books and theses | d). Preparing research articles

29. For a proposal, the abstract generally uses [A] / [CO4]

a). Future-oriented language | b). Past tense only B) | c). Present tense only | d). Passive voice only

30. A major advantage of LaTeX is [An] / [CO4]

a). Easy photo editing and graphic design | b). Automatic collection of experimental data

c). Direct measurement of physical quantities | d). High-quality typesetting

[1x30 = 30]

Part B

Essay type question.

Answer all questions

Each question carries **20** marks

31. Prepare a professional Curriculum Vitae (CV) using LaTeX for a final-year undergraduate student applying for a research internship. Your CV should contain: (a) personal and contact information, (b) career objective/profile, (c) educational qualifications, (d) technical and scientific skills, (e) projects/research experience, (f) achievements, certifications and extracurricular activities, and (g) references/contact details where appropriate. Write the appropriate LaTeX commands and structure required to create the CV. Explain how LaTeX helps maintain consistent formatting and professional presentation. [A] / [CO4]

[20x1 = 20]